Lecture for 9th week

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Example 4

QUESTION 3 (20Pts): A jib crane has the dimensions shown in the Figure 3. Assume that the rod AB and tube BC have the **same Young's modulus** E and **same cross-sectional area** A. Determine the **vertical displacement** at point B caused by the application of **force** P.

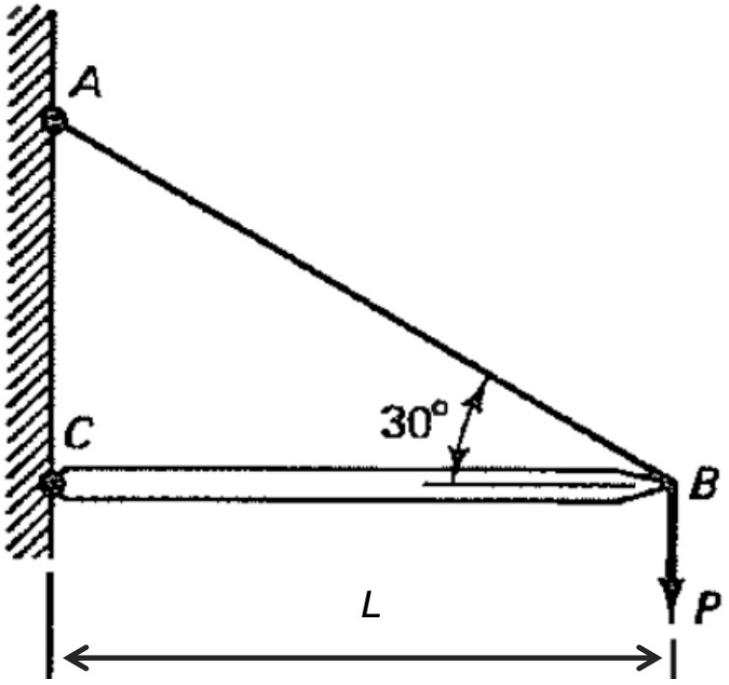
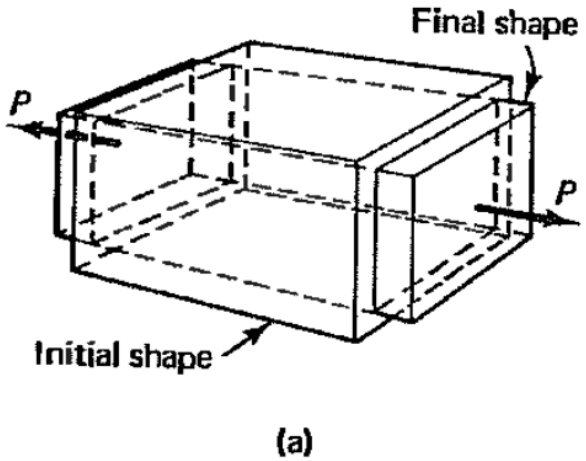


Figure 3

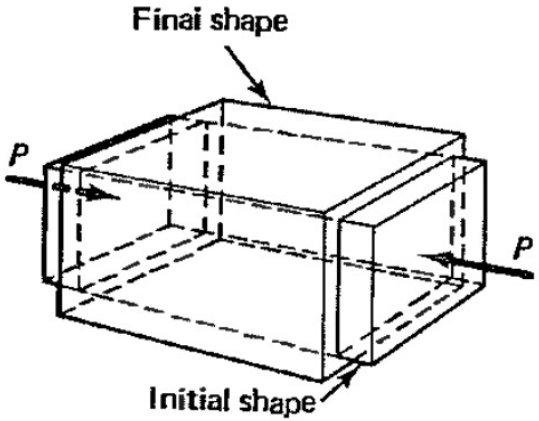
1. Poisson's Ratio

- If a solid body is subjected to an axial tension/compression, it contracts/expands laterally.



(a)

lateral contraction



(b)

lateral expansion

Poisson's Ratio

$$\nu = \left| \frac{\text{lateral strain}}{\text{axial strain}} \right| = -\frac{\text{lateral strain}}{\text{axial strain}}$$

It is a constant for elastic, homogeneous, isotropic materials, and is a definite property of a material.

2. Thermal Strain and Deformation

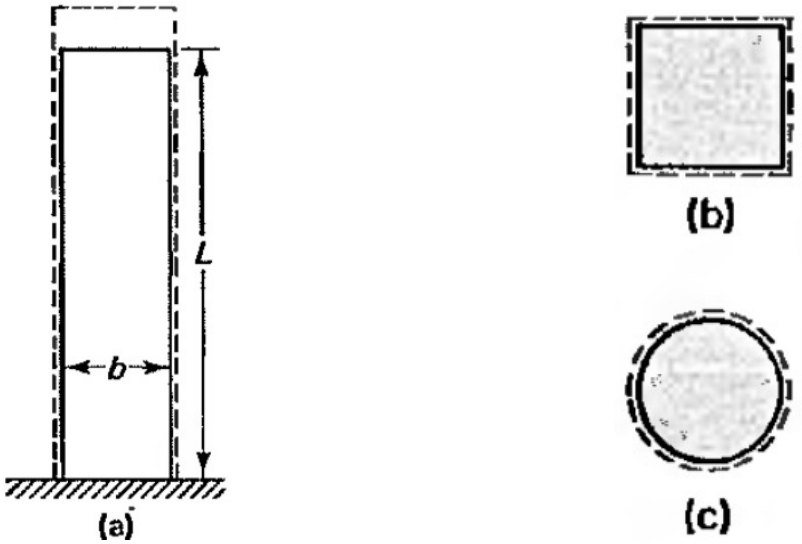
With changes in temperature, solid bodies expand on increase of temperature and contract on its decrease. The thermal strain ε_T caused by a change in temperature from T_0 to T expressed as

$$\varepsilon_T = \alpha(T - T_o)$$

over a limited range of temperature change, α is experimentally determined thermal coefficient of thermal expansion constant.

2. Thermal Strain and Deformation

Equal thermal strains develop in every direction for unconstrained homogeneous isotropic materials.

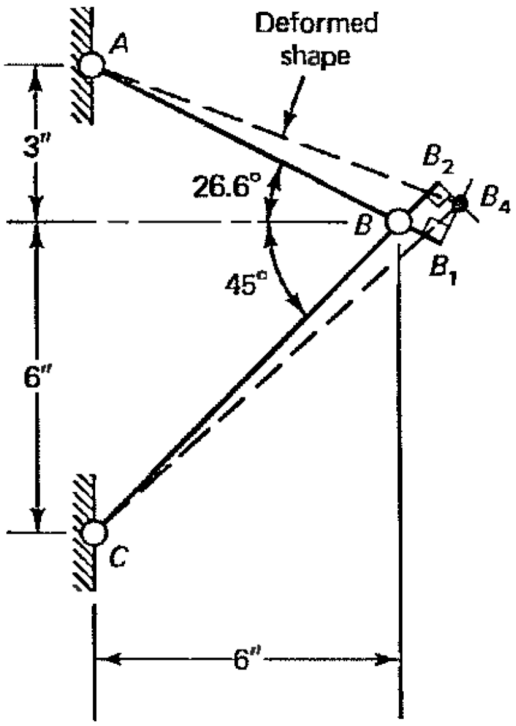


thermal deformation

$$\Delta_T = \alpha(\delta T)L$$

3. Thermal Strain and Deformation

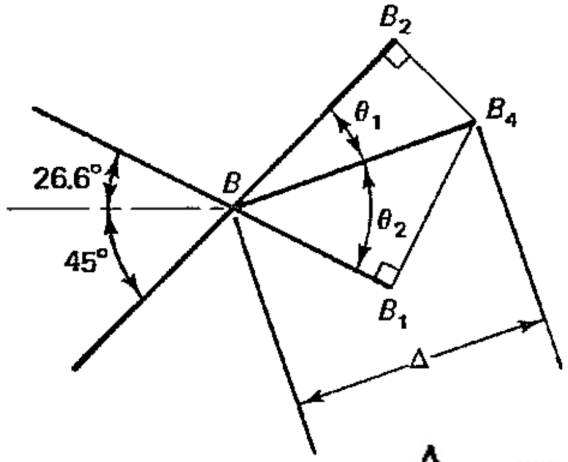
Determine the displacement of point B in Example 2-4 caused by an increase in temperature of 100°F. See Fig. 2-29(a). the coefficient of thermal expansion for 2024-T4 aluminum alloy is 12.9×10^{-6} per °F.



(a)

$$\Delta_{AB} = 12.9 \times 10^{-6} \times 100 \times 6.71$$

$$\Delta_{BC} = 12.9 \times 10^{-6} \times 100 \times 8.49$$



(b)

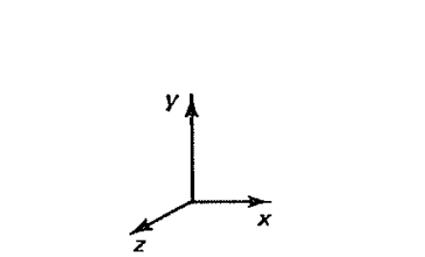
$$\Delta_T = \Delta_{BC}/\cos\theta_1 = 12.2 \times 10^{-3} \text{ in}$$

forming an angle of $45° - \theta_1 = 18.4°$ with the horizontal.

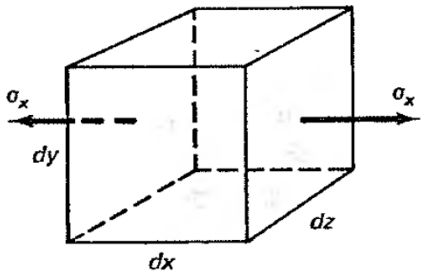
4. Elastic strain energy for uniaxial stress

In solid deformable bodies, stress multiplied by their respective areas are forces, and deformations are distances. The product of these two quantities is the internal work done in a body by externally applied forces. This internal work is stored in an elastic body as the internal elastic energy of deformation, or the elastic strain energy.

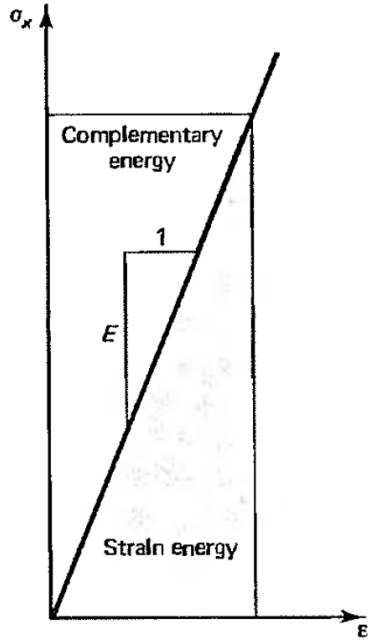
4. Elastic strain energy for uniaxial stress



linear elasticity



(a)



(b)

force $\sigma_x\, dy\, dz$ gradually increases from 0 to final value

$$dU = \underbrace{\tfrac{1}{2}\sigma_x\, dy\, dz}_{\text{average force}} \times \underbrace{\varepsilon_x\, dx}_{\text{distance}} = \underbrace{\tfrac{1}{2}\sigma_x\varepsilon_x\, dx\, dy\, dz}_{\text{work}} = \tfrac{1}{2}\sigma_x\varepsilon_x\, dV$$

strain energy density

$$U_o = \frac{dU}{dV} = \frac{\sigma_x\varepsilon_x}{2}$$

$$U_o = \frac{dU}{dV} = \frac{E\varepsilon_x^2}{2} = \frac{\sigma_x^2}{2E}$$

$$U = \int_{\text{vol}} \frac{\sigma_x^2}{2E}\, dV$$

an element in uniaxial tension (a) and a linear stress-strain diagram.

4. Deflection by the Energy Method

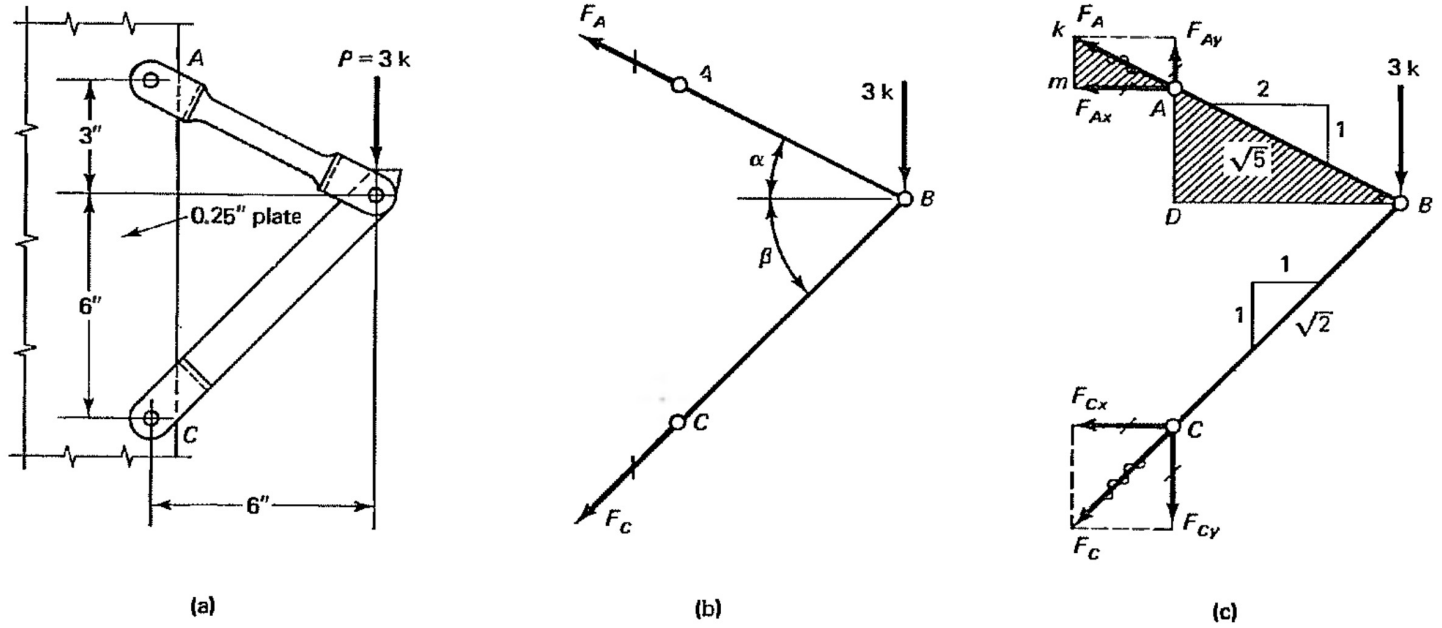
The principal of energy conservation may be very effectively used for finding deflections (变形量) of elastic members due to applied forces. Here a more limited objective, determining the deflection caused by the application of a single axial force, is considered. Thus, internal strain energy is equated to external work of applied force

$$U = W_e$$

Assuming the external force is applied gradually.

4. Elastic strain energy for uniaxial stress

For the bracket analyzed for stresses in Example 1-3, determine the deflection of point B caused by the applied vertical force $P = 3$ kips. Also determine the vertical stiffness of the bracket at B. Assume that the members are made of 2024-T4 aluminum alloy and that they have constant cross-sectional areas, i.e., neglect the enlargements at the connections. See idealization in Fig. 2-24(a).



4. Elastic strain energy for uniaxial stress

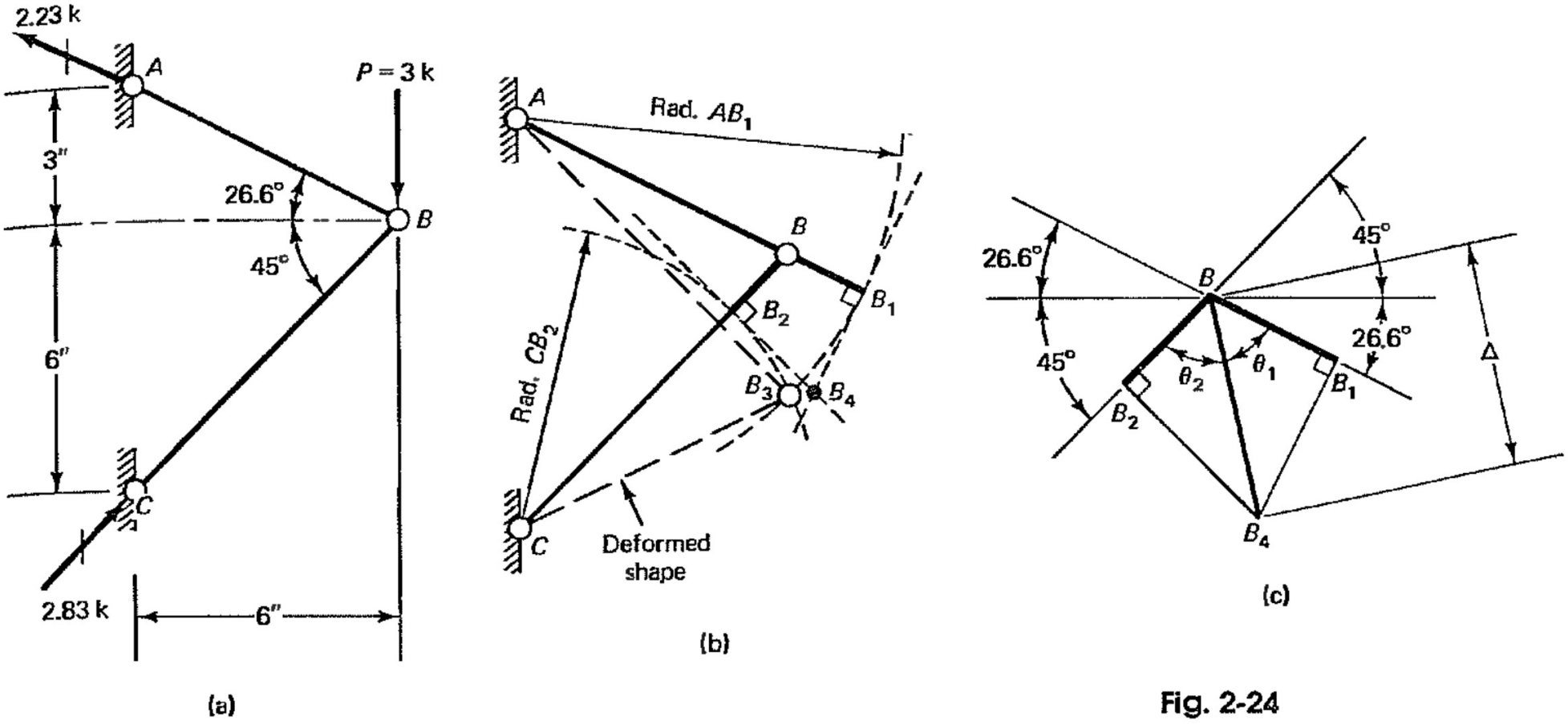


Fig. 2-24

4. Elastic strain energy for uniaxial stress

$$\Delta_{AB} = \left[\frac{PL}{AE}\right]_{AB} = \left[\sigma\frac{L}{E}\right]_{AB} = \frac{17.8 \times 6.71}{10.6 \times 10^3} = 11.3 \times 10^{-3} \text{ in}$$

(elongation)

$$\Delta_{BC} = -\frac{12.9 \times 8.29}{10.6 \times 10^3} = -10.3 \times 10^{-3} \text{ in} \qquad \text{(contraction)}$$

$$\Delta_{\text{vert}} = \Delta \cos 11.2^\circ = 18.0 \times 10^{-3} \text{ in}$$

$$U = \frac{1}{2}\frac{P_{AB}^2 L_{AB}}{A_{AB}E} + \frac{1}{2}\frac{P_{BC}^2 L_{BC}}{A_{BC}E} = \frac{1}{2}P\Delta_{\text{vert}}$$